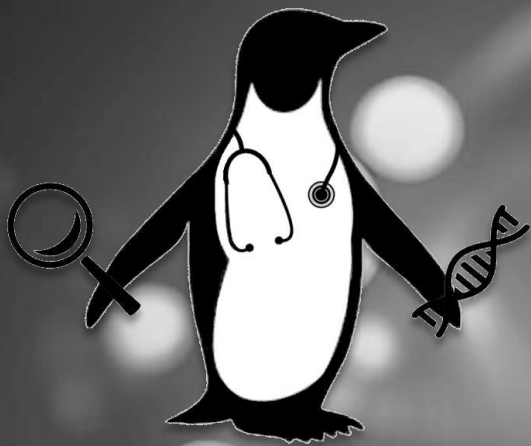




PENGQUIN Genomic Context

VITO Co-promoters: Dr. Bart Buelens + Dr. Gökhan Ertaylan
UGent Promoters: Dr. Ruben Taelman + Dr. Ruben Verborgh

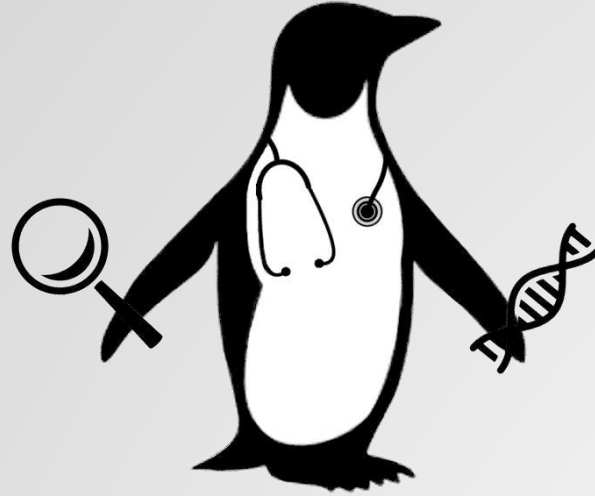
Elias Crum

Agenda

1. PENGQUIN Project Introduction
2. Introduction to Genome Sequencing
3. Genome Data
4. Current Genomic Data Storage and Querying Strategies
5. Genomic Data as Linked Data?

Things to note

1. If you have a question at any point, PLEASE ASK!!
2. I will ask “does this make sense” – I mean it (not just rhetorically)
3. This presentation is VERY HIGH LEVEL



PENGQUIN

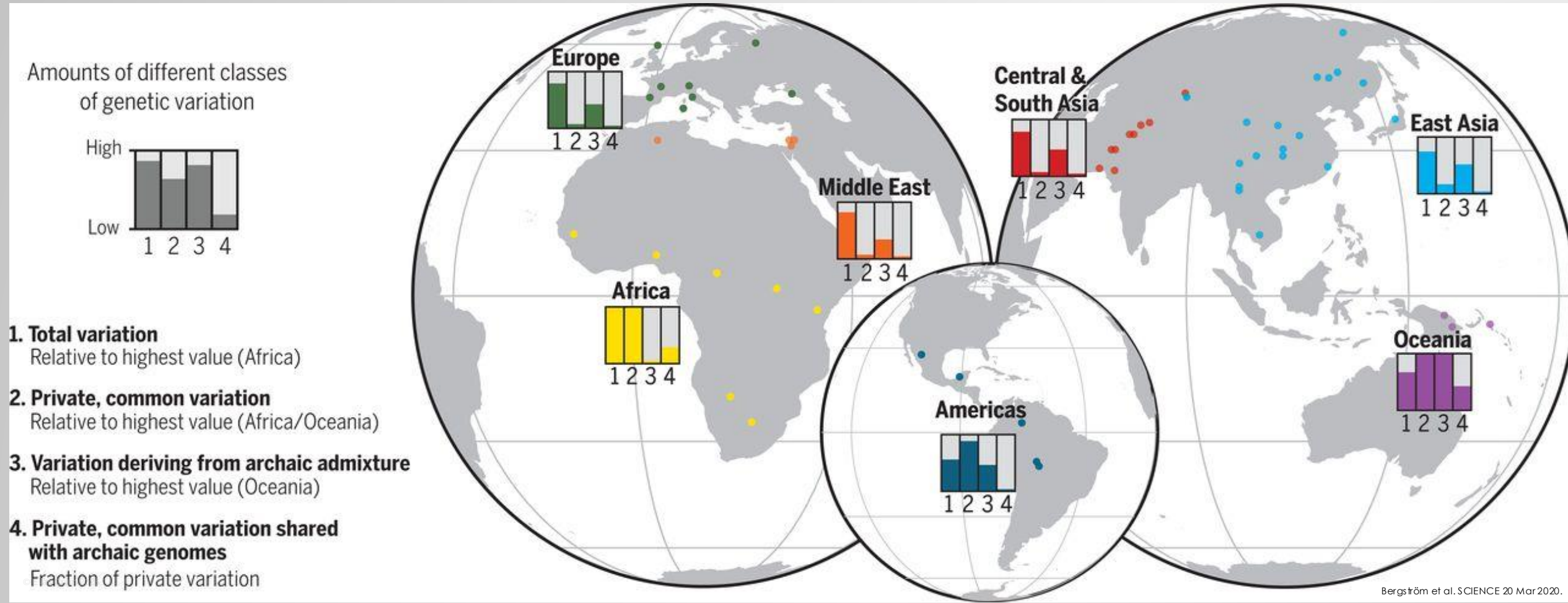
PErsoNal Genome QUery IN healthcare and clinical practice

Human Genome Variation

Global genetic variation (929 genomes)

- 67.3 million single-nucleotide polymorphisms
- 8.8 million small insertions or deletions (indels)
- 40,736 copy number variants

All humans are 99.9% genetically identical



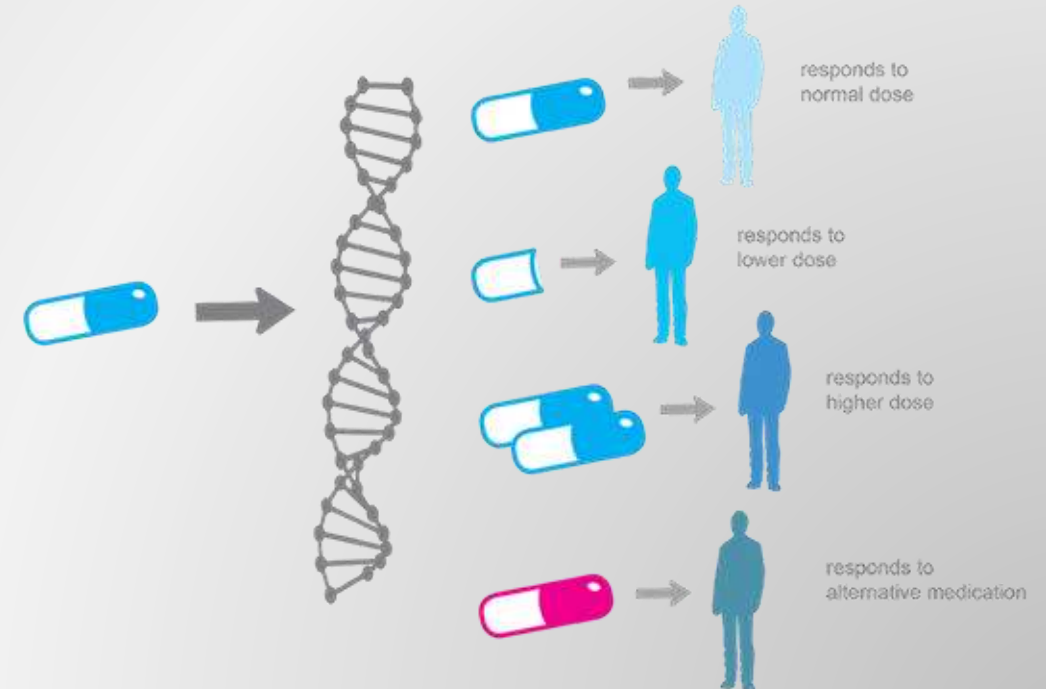
Personalized Medicine

All human diseases are influenced by genetics

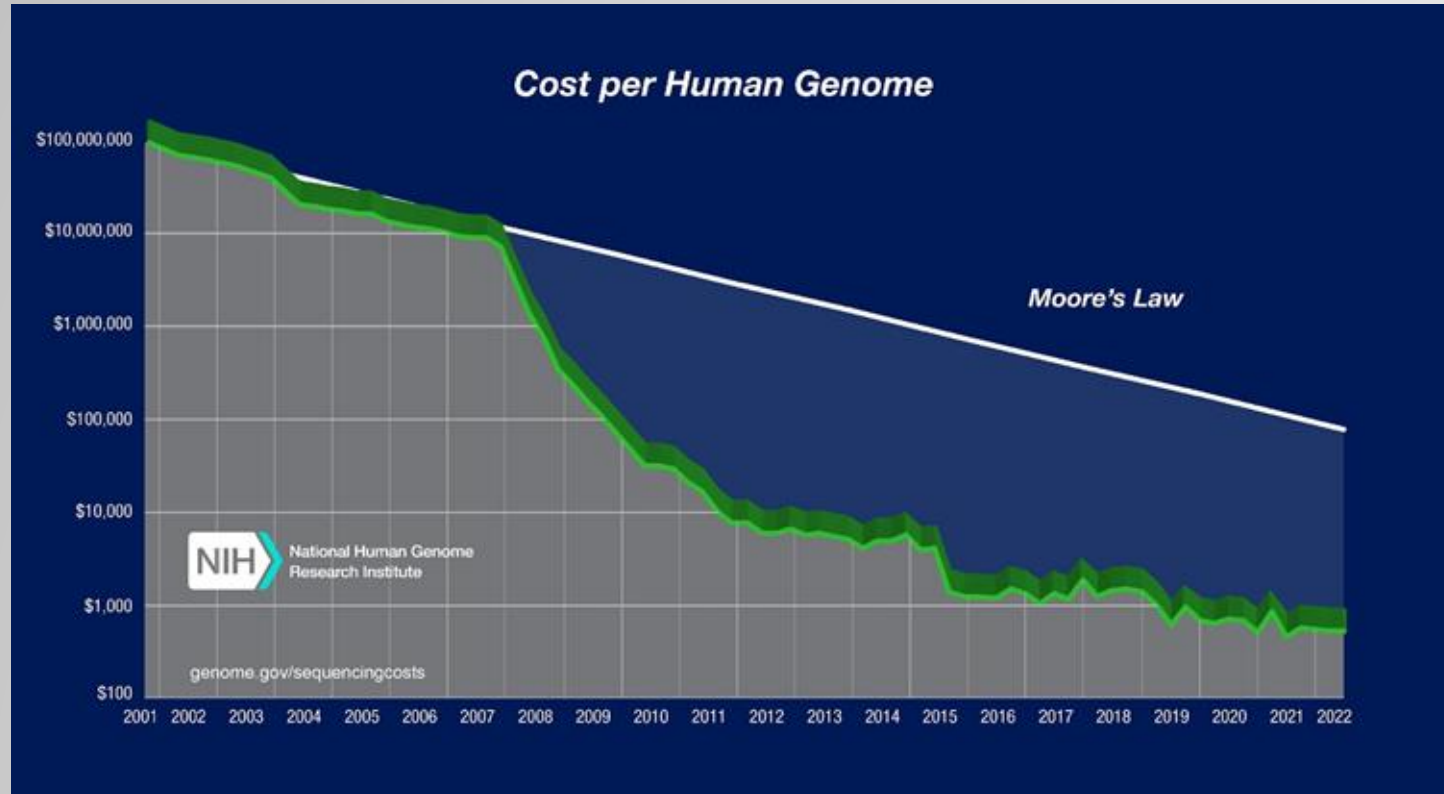
Personalized Medicine = Medical care informed by an individual's genetic profile

Examples:

- Drug prescription
- Cancer diagnosis
- Cancer therapy development and targeting
- Genetic disease screening



Sequencing Costs



<https://www.genome.gov/about-genomics/fact-sheets/Sequencing-Human-Genome-cost>

Currently ~€300 per genome

Per-base cost of **sequencing** has been dropping by half every ~5 months

Research Context

Challenges to personalized medicine?



Genome storage and querying



Why?



Privacy concerns



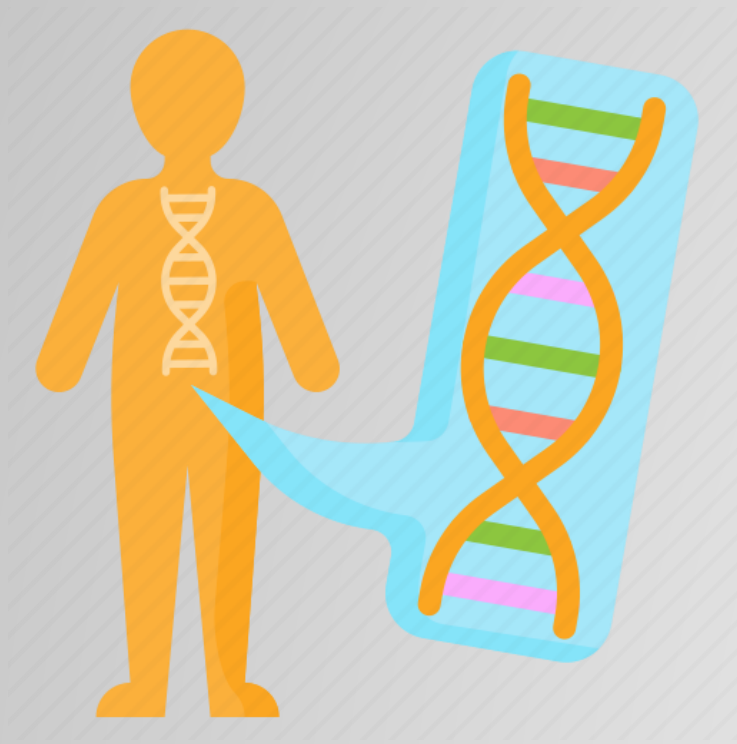
Genome storage

~3 BILLION
base pairs



Stored data querying

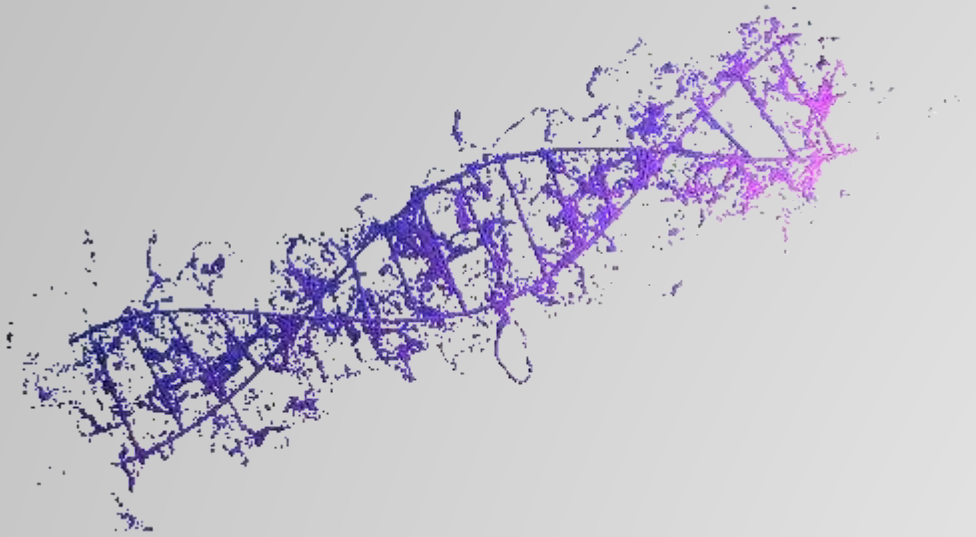
How do we go from:



```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)
ACCCCTCTTTTCTTATCATTGACATTTAAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCGGGCTCCGGCCCCGGCCCGGCTCGGGGCCCGCGGGGCCCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCGCCCCAAGTGGCCCCGGGGCTTGATTTTTGCTTTTAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGAAGTGTCTT
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGGTCGGGCGAGGTGCAGCAGAAGTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

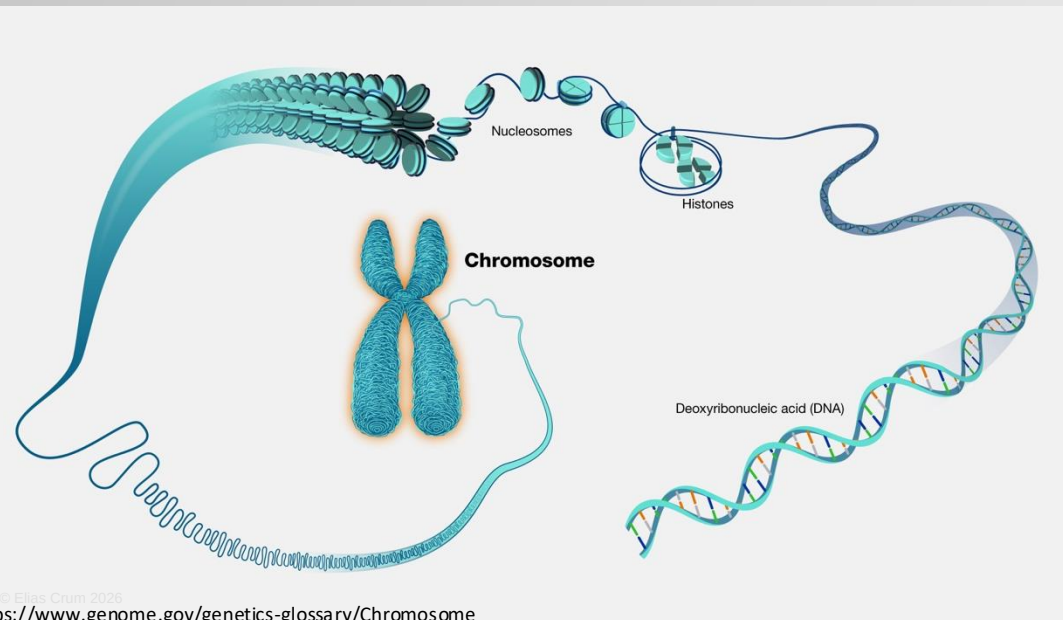
To this?

The VERY basics:



Deoxyribonucleic Acid (DNA):

- 2 strands (that are complementary to each other)
- At each position contains a nucleotide (base)
 - A, T, C, G
 - A-T / C-G
- Wound tightly together into chromosomes
- Within EVERY living cell of your body (other than red blood cells)
- Each human has $\sim 108 \times 10^9$ **kilometers** of DNA



Genome Sequencing

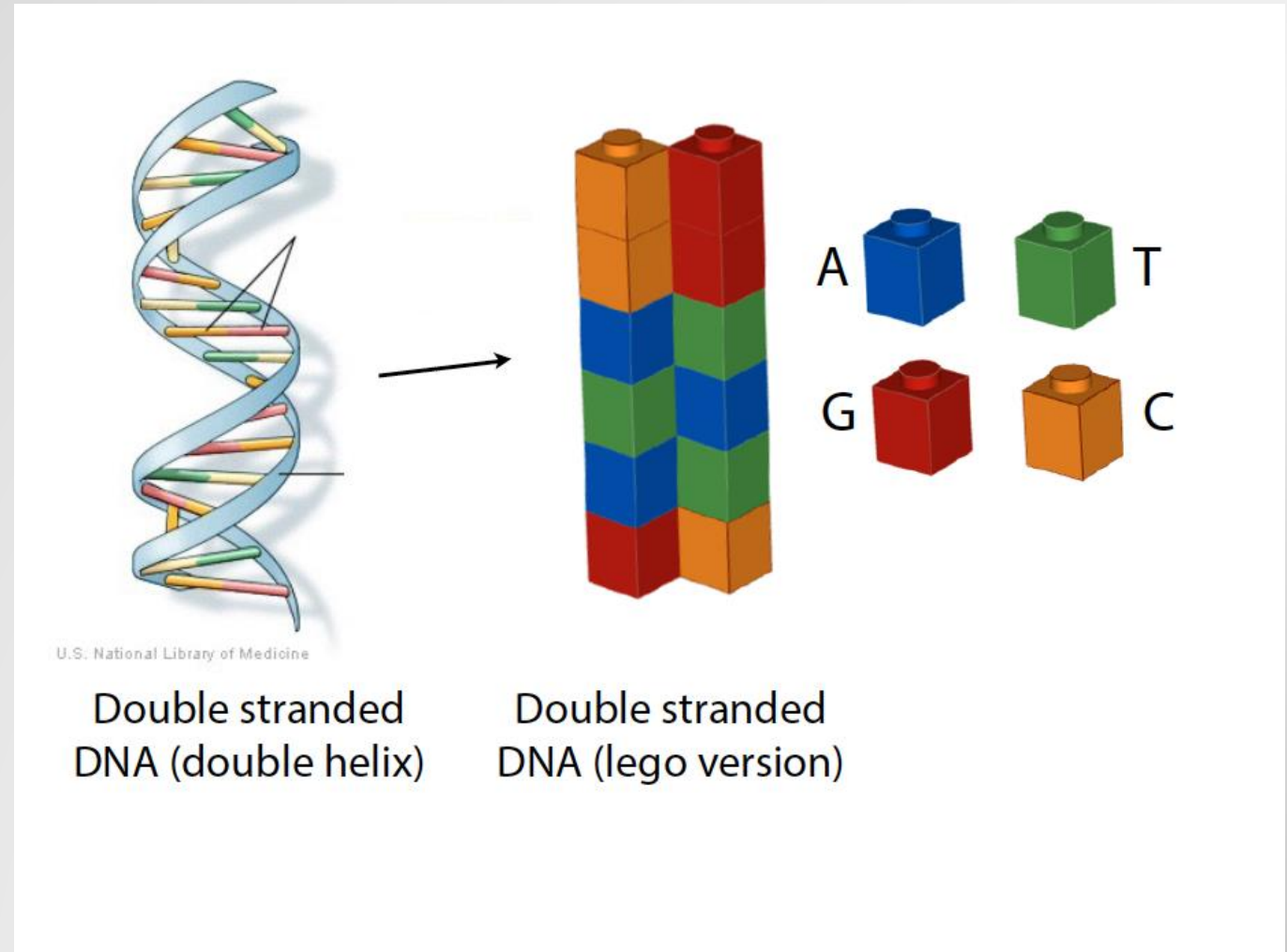
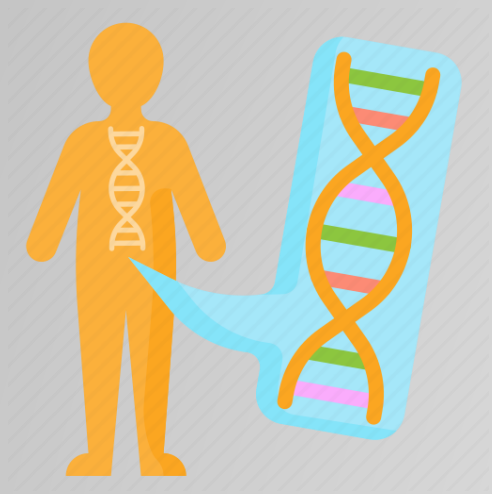


```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)
ACCCTCTTTCTTATCATTGACATTTAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCGGGCTCCGGCCCCGGCCCGGCTCGGGGCCCGCGGGCCCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCGCCCCAAGTGGCCCCGGGGCTTGATTTTGTCTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGTGGAGGAGGACTTGTCTT
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGCGGGAGTCTGCTGCTGCTGCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGGTCGGGCGAGGTGCAGCAGAAGTTCCC
GCGGTTGCAAAAGTCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

Three Generations of Sequencing Technologies:

- 1st Gen: Sanger Sequencing (~2000s)
- 2nd Gen: Shotgun Sequencing (~2005 – Current)
- 3rd Gen: Long-read Sequencing (2015 – Current)

Quick Genome Sequencing Overview



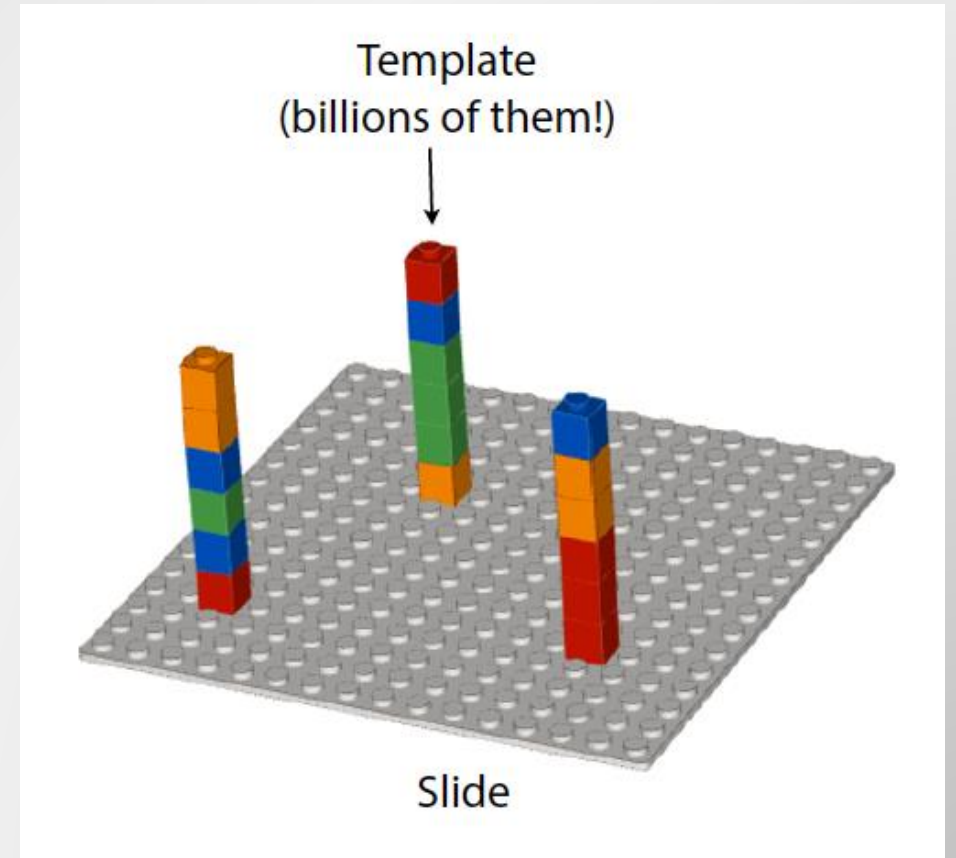
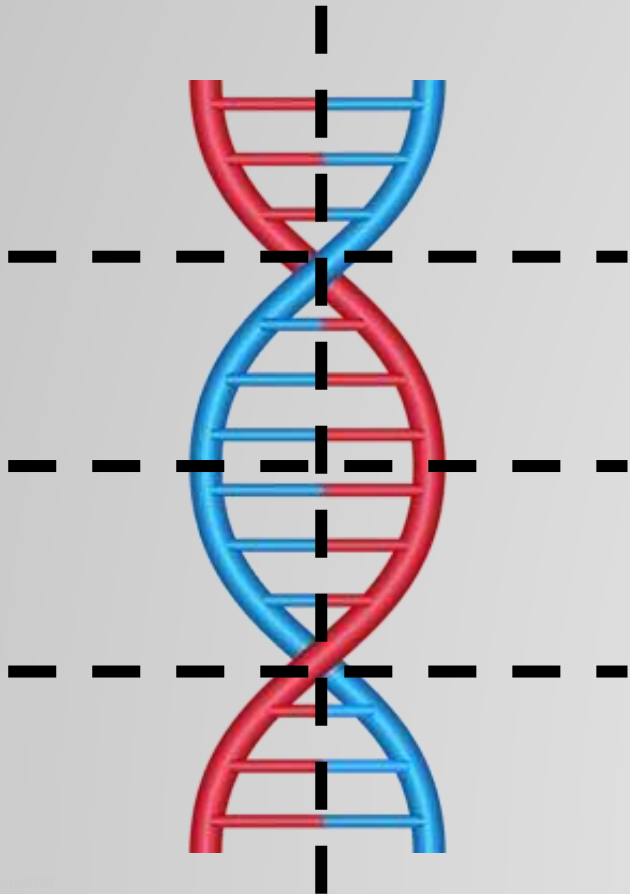
Most cost-effective strategy currently:

- 2nd Gen Sequencing
 - Short Reads (~150 base pairs)
 - High Throughput (>30x representation of each base in genome)

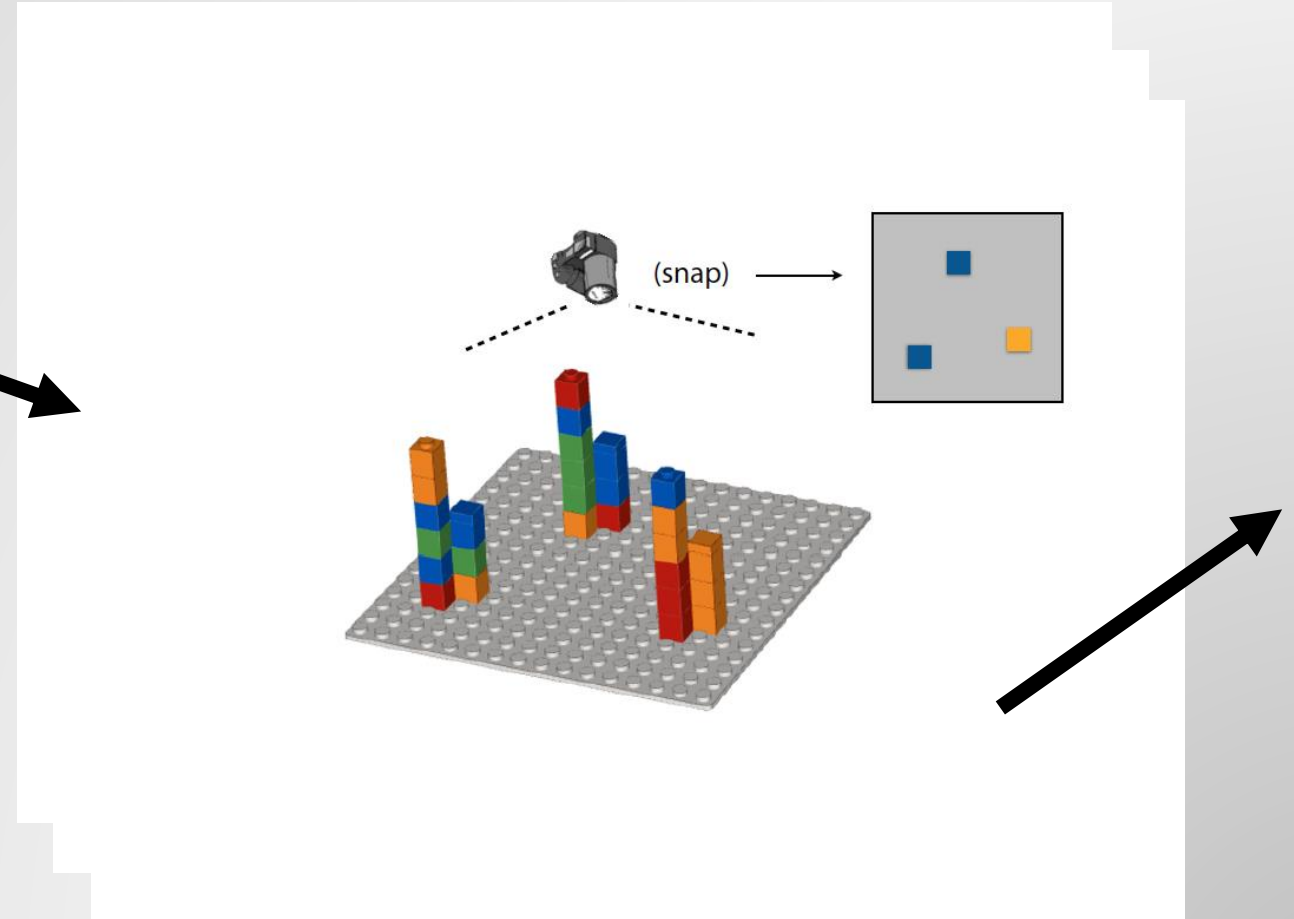
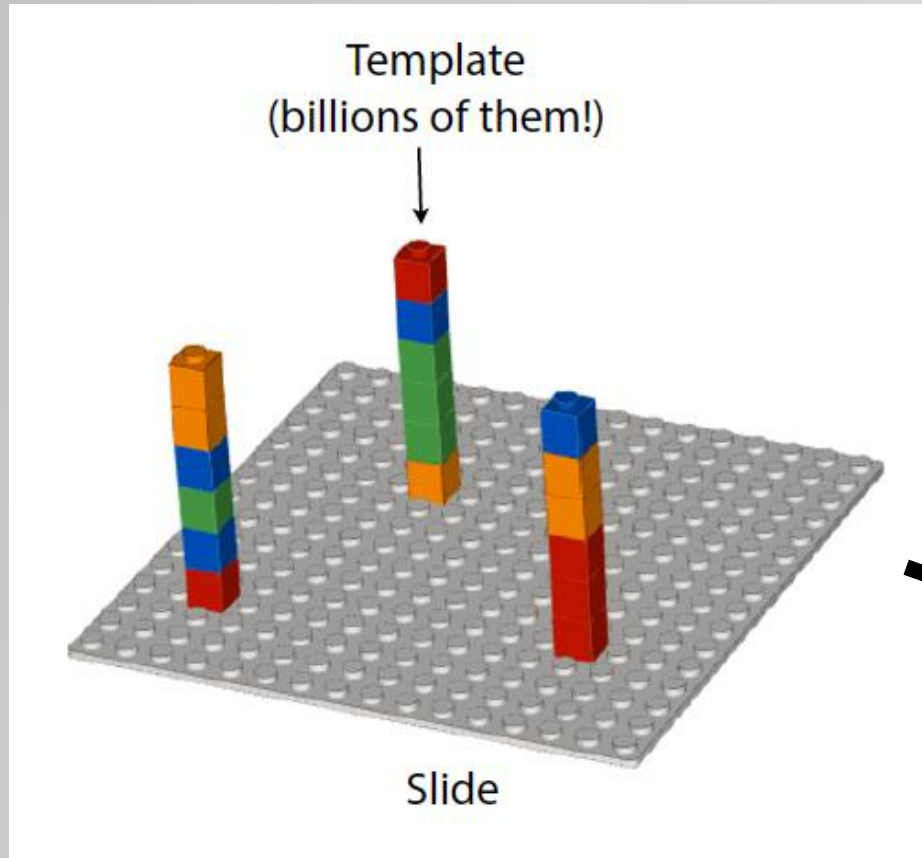
2nd Gen Sequencing Preprocessing

** Most cost-effective strategy currently:

Short Reads (~100 - 150 base pairs)



Quick Genome Sequencing Overview



Sequencing by synthesis

Cycle 1 Cycle 2 Cycle 3 Cycle 4 Cycle 5 Cycle 6

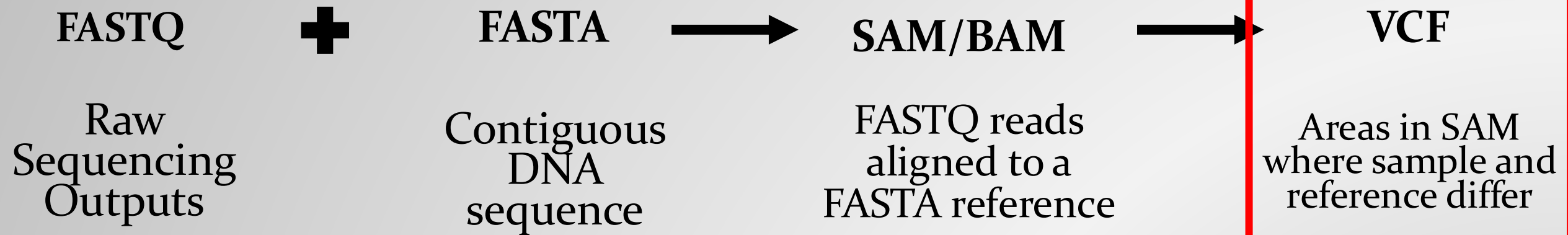
complement complement complement complement complement complement

G A T A C C

CCATAG

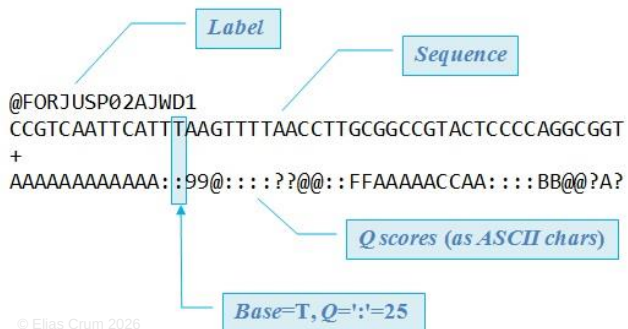
#K00282:309:HWT23BBXX:4:1101:1610:1086
TGGGAGTGTCTTCAGCCCTAAGGCAGCATCAGAGATCATAATCTTCTGCTC
+
AAFFJ-AJFJFFJFJFFJ<JJAFJJFFJJJJAF<AJFJFJFF<JJFJF<
@K00282:309:HWT23BBXX:4:1101:1671:1086
CCAGAAGGAGGCAGAGCTCCGCACCTGGATCGAGGGACTCACCGGCCTCTC
+
-<AFFJJAJF<AJJFJJJ<FJJJ7FJJJJ<JJJJJJFJFJJF<-FJJJJFJ
@K00282:309:HWT23BBXX:4:1101:2179:1086
GGGCACTCGGTACTGCTGCAGGAGCTGTTGCTGGAAGTGGAACCCAGGCGG
+
--AAFJJFAJA7A-F<-FJF<AAAJFF--F-7FJ-<FFFJ<-A777AFAFF
@K00282:309:HWT23BBXX:4:1101:2402:1086
ACCATCACCTCGAGGTTGAACCCTCGGATACGATAGAAAATGTAAAGGCC
+
AAFFJJJJFJJFFJJJJJJJJJJJJJJJJJJJJFFJJFJJ-FJJJJFJJJ
@K00282:309:HWT23BBXX:4:1101:2625:1086
CGGAGACAGGGCAGACTGTGCCATCTCCTCTCCTCACCAGCTGTGTACCT
+
<AF-F-FAFJJF<J<<A<AAJ7FJJJJJJJJJJ<J-<<---AAAJJJJJFJ7

Sequence Data Formats

[illegible]

```
##fileformat=WCFv4.3
##fileDate=20090805
##source=msiMutationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=ID=20,length=62435964,assembly=B36,md5=f126c428a603f794618ff6bde2ba,species="Homo sapiens",taxonomy=x
##phasing=partial
##INFO=
##INFO=
##INFO=
##INFO=
##INFO=
##INFO=
##FILTER=
##FILTER=
##FORMAT=
##FORMAT=
##FORMAT=
##FORMAT=
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA00001 NA00002 NA00003
20 14370 ref6054267 G A 29 PASS NS-3,DP:14,AF=0.5;DB;H2 GT;GQ;DP;HQ 10:048:1,81,51 11:048:8,51,51 1:1/43:5;...
20 17330 T A 3 g10 PASS NS-3,DP:11,AF=0.017 GT;GQ;DP;HQ 10:048:3,58,50 11:048:3:5;65,3 0/0:41:1
20 1110696 ref6040355 T A 29 PASS NS-3,DP:10,AF=0.333,0.667;AA=T;DB GT;GQ;DP;HQ 12:21:6,0,23,27 21:2:19,2 2/2:36:4
20 1230237 T A 47 PASS NS-3,DP:13,AA=T GT;GQ;DP;HQ 0154:75,66,60 10:048:4,51,51 0/0:61:2
20 1234567 microsat GTC G,GTCT G PASS NS-3,DP:9,AA=G GT;GQ;DP;HQ 014:15:4 0/2:17:2 1/1:40:3
```

```
@#HD VN:1.0 SO:coordinate
@SQ SN:chr20 LN:64444167
@PG ID:-topHat VN:2.0.14 CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-read-lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
HMI-ST1145:74:C10IDACXX:7:1102:4284:73714 16 chr20 190939 3 100M * 0 0
CCGTGTTTAAAGGTGGATCGGCCTCACCTTCCCAGCATGGCTTACTAGTGCCCTAGGAATCCAGTAGTACTGCTGCTCTCAGTCCCCCTCT
C BBDDCCDCCDDDDDDDDDDCCDCBBD?DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDBHHFFFD@@
AS:i:-15 XM:i:3 XO:i:0 XG:i:0 MD:Z:55C20C1A9 NM:i:3 NH:i:2 CC:Z= CP:i:55352714 HI:i:0
HMI-ST1145:74:C10IDACXX:7:1114:2759:41961 16 chr20 193953 50 100M * 0 0
TGCTGATCATCTGGTATTAGTGCTTCTGACTCAGAAGACCTCTGCTCCCTGGGGCAGTGACCACTTCAGTGATTCCTCCGACATAAGGGGCATGGACGA
G CDDDDDEDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDEEC=DFFEJJJJJJJIGJHBJJJJJJJJJJJJJJJJJJJJJJJHHHHHHFFFFCCC
AS:i:-16 XM:i:3 XO:i:0 XG:i:0 MD:Z:6GB16T1873 NM:i:3 NH:i:1
HMI-ST1145:74:C10IDACXX:7:1204:14760:4030 16 chr20 276877 50 100M * 0 0
GGCTTTATTGGTA AAAAGGAATAGCAGATTATGACAATAATCCCACTGGCCGACAGCCAACCAAGAAAGGAAGGAAGAAGCAGGAAAAACCA
C DDDDDDDDDDDDDDDDDDDDDDEEEFEFFEGHHHFGDJJIHJIIJJJJIIIGFJJIHIIIIJJJJJJJJIGHFAHGHJFHFGHGFDDGBB
AS:i:-11 XM:i:2 XO:i:0 XG:i:0 MD:Z:0A8513 NM:i:2 NH:i:1
HMI-ST1145:74:C10IDACXX:7:1210:11167:8699 0 chr20 271218 50 50M4700N50M * 0 0
0 GTGGCTTCCACAGGAATGTTGAGGATGACATCCATGCTGGGGTGACCTGGGTCTCCGAGCAGAACATCTCCAATATGACCTCTCG
```



Sequencing Storage



The human genome = ~3 billion base pairs

- 3 billion letters -- ~700 megabytes

Representative of a **single strand** of the whole human genome sequence

E.g. : ...ATGCCGTAAACAGATGTCA ...

Format: *TXT* file

- Raw human whole genome sequence -- ~200 gigabytes

Whole genome sequences typically have **>=30x coverage at each base position**. (i.e. each base of the genome is represented in the sequence file a ≥ 30 times)

Format: *FASTQ* file

- Aligned human whole genome sequence -- ~210 gigabytes // ~100 gigabytes // ~15 gigabytes

Whole genome sequence **is mapped to a reference human genome** and can be stored in various formats.

Recent advances have allowed for improved compression.

Format: *SAM* file // *BAM* file // *CRAM* file

- Human genome variant file (list of mutations) -- ~125 megabytes

Individual humans only exhibit **~0.1% variation** comparatively (~3 million mutations). Variant file **records those differences** in one individual's genome compared to the “normal” reference genome.

Format: *VCF* file

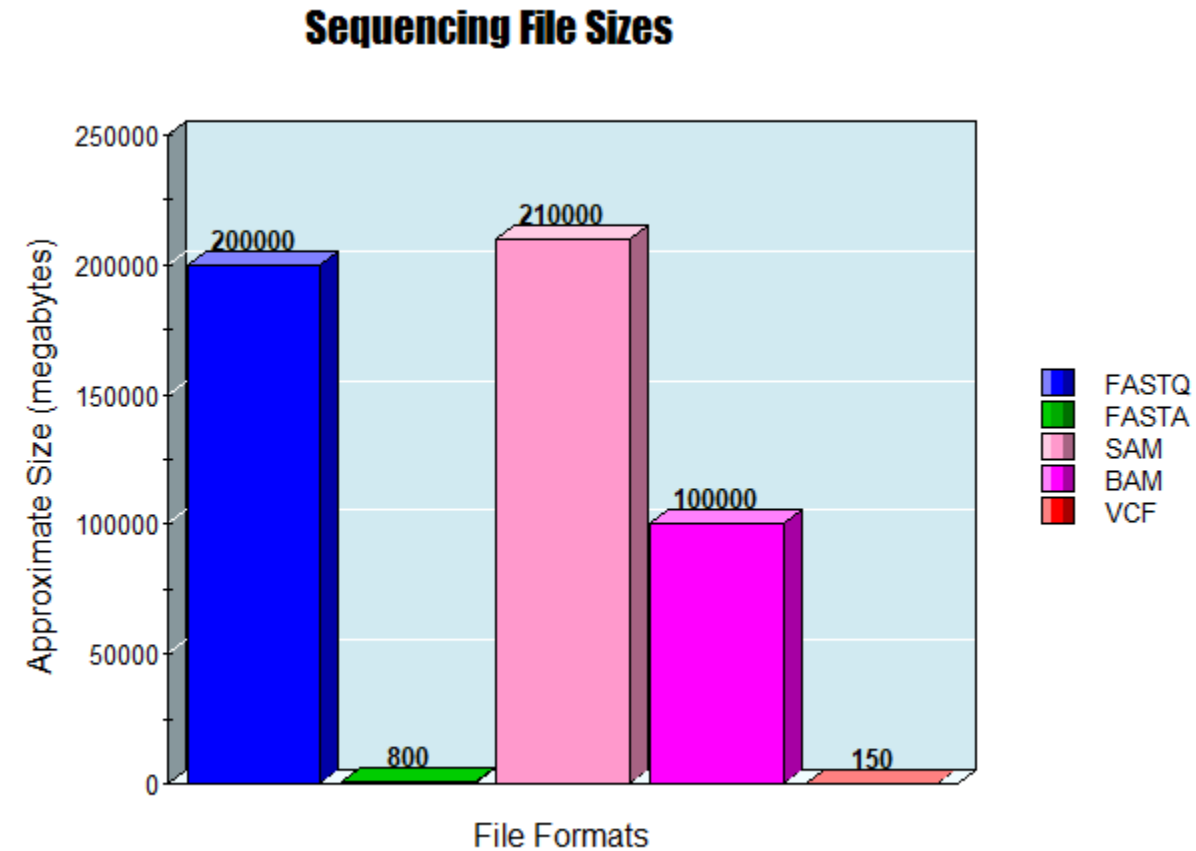
Sequencing File Sizes

FASTQ ~ 200 **gigabytes**

FASTA ~ 800 **megabytes**

SAM/BAM ~ 210/100 **gigabytes**

VCF ~ 150 **megabytes**



Variant Call Format (VCF)

- tab delimited .txt file
- Rows represent individual mutations in a sequence (when compared to reference)
- Columns store information about that mutation

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT
--------	-----	----	-----	-----	------	--------	------	--------



```
##fileformat=VCFv4.1
##fileDate=20140930
##source=23andme2vcf.pl https://github.com/arrogantrobot/23
##reference=file:///23andme_v3_hg19_ref.txt.gz
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT GEN
chr1 82154 rs4477212 a . . . . GT 0/0
chr1 752566 rs3094315 g A . . . . GT 1/1
chr1 752721 rs3131972 A G . . . . GT 1/1
chr1 798959 rs11240777 g . . . . GT 0/0
chr1 800007 rs6681049 T C . . . . GT 1/1
chr1 838555 rs4970383 c . . . . GT 0/0
chr1 846808 rs4475691 C . . . . GT 0/0
chr1 854250 rs7537756 A . . . . GT 0/0
chr1 861808 rs13302982 A G . . . . GT 1/1
chr1 873558 rs1110052 G T . . . . GT 1/1
chr1 882033 rs2272756 G A . . . . GT 0/1
chr1 888659 rs3748597 T C . . . . GT 1/1
chr1 891945 rs13303106 A G . . . . GT 0/1
```


Variant Call Format (VCF)

* Important note: 1 VCF file CAN represent mutations of MANY sequences

```
##fileformat=VCFv4.2
##fileDate=20151002
##source=callMomV0.2
##reference=gi|251831106|ref|NC_012920.1| Homo sapiens mitochondrion, complete genome
##contig=<ID=MT,length=16569,assembly=b37>
##INFO=<ID=VT,Number=.,Type=String,Description="Alternate allele type. S=SNP, M=MNP, I=Indel">
##INFO=<ID=AC,Number=.,Type=Integer,Description="Alternate allele counts, comma delimited when multiple">
##FILTER=<ID=fa,Description="Genotypes called from fasta file">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT HG00096 HG00097 HG00099 HG00100 HG00101 HG00102 HG00103 HG00105 HG00106 HG00107
MT 10 . T C 100 fa VT=S;AC=3 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 16 . A T 100 fa VT=S;AC=3 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 26 . C T 100 fa VT=S;AC=3 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 35 . G A 100 fa VT=S;AC=2 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 40 . TC CT 100 fa VT=M;AC=1 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 41 . C T 100 fa VT=S;AC=4 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 42 . TCC CCC,T 100 fa VT=S,I;AC=1,1 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 46 . T C 100 fa VT=S;AC=1 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 47 . G A 100 fa VT=S;AC=1 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 52 . TGG CAA 100 fa VT=M;AC=1 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 55 . TATTTT T,CATTTT,AATTTT,TTT,TTTTT 100 fa VT=I,S,S,I,I;AC=5,3,2,1,1 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 57 . T C 100 fa VT=S;AC=3 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 58 . TTT T 100 fa VT=I;AC=4 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
MT 59 . T A 100 fa VT=S;AC=1 GT 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

Existing VCF Querying Tools?

- VCFio
- Seqminer2
- BCFtools

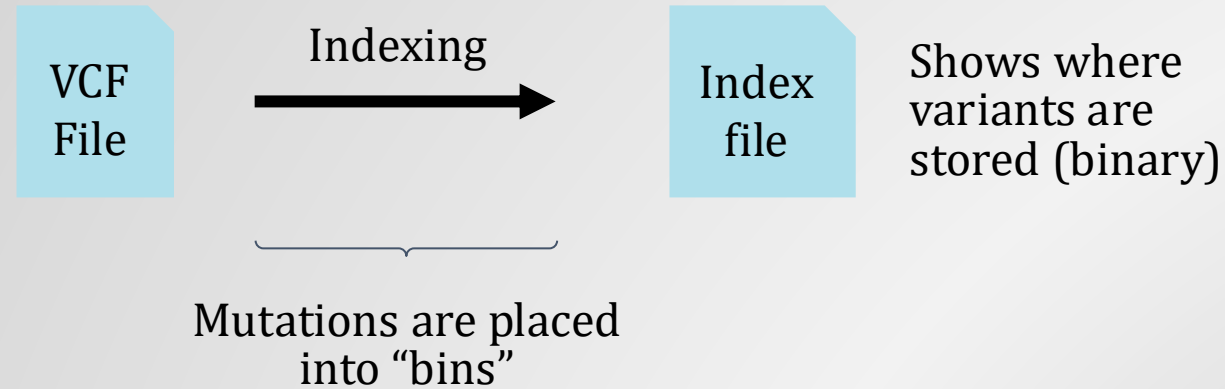
Tabix --> hybrid of binning and linear index

Made to query 1.000+ VCF (and BCF) files all stored in the same location

Small VCF analysis can mostly be performed by brute-force parsing

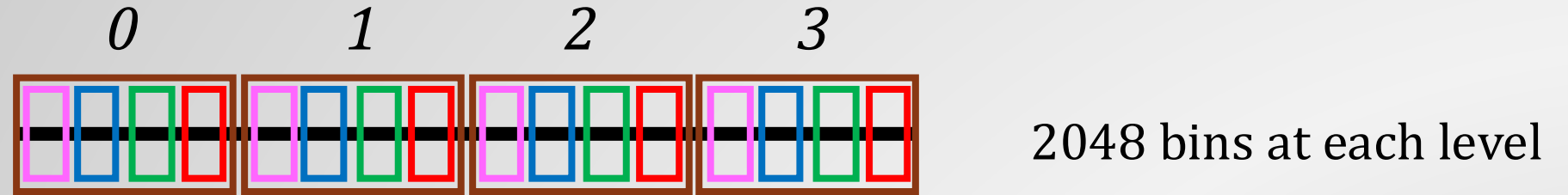
Existing VCF Querying Tools?

Tabix Strategy:



A "bin" : Represents a subset of byte locations within a VCF file == Range of base positions on a chromosome

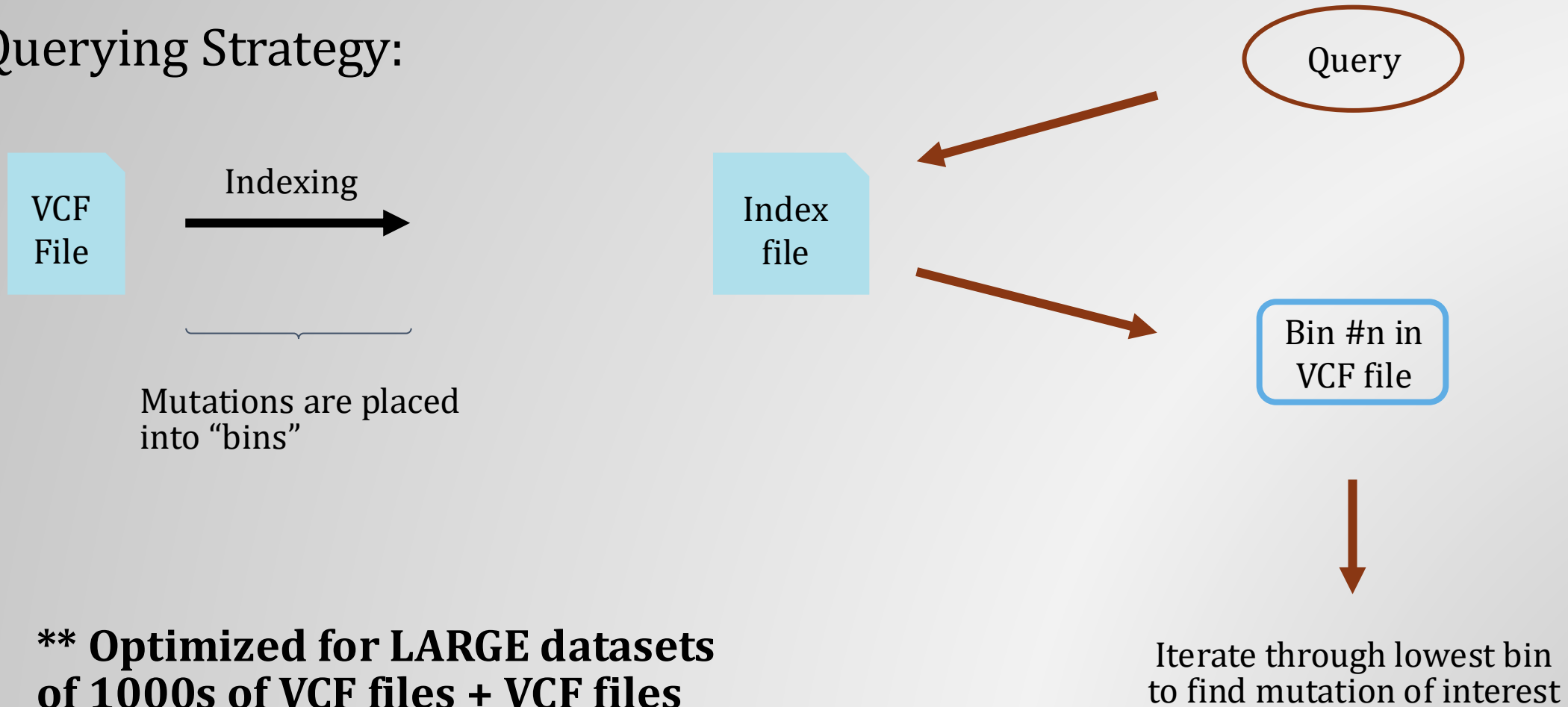
VCF File:



1. Bins have a hierarchy
2. Bins contain a subset of mutations represented in the VCF file
3. Bin memory size is determined by file size
4. Index contains information about mutation byte offset
→ specific look-up coordinates
5. Typically, VCF files are .gz compressed

Existing VCF Querying Tools?

Querying Strategy:



**** Optimized for LARGE datasets
of 1000s of VCF files + VCF files
representing MANY sequences ****

What if we used linked data strategies?

Architecture of data storage?

What will the links be?

How can we leverage the different architecture?

Query engines? And how will data transfer be handled?

Previous Implementation

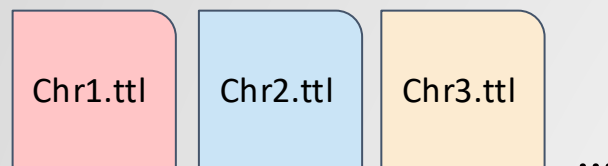


VCF
File

Download + decompress
vcf.gz file

**** Serialization + Querying takes ~30 minutes ****

For more detail see https://github.com/ecrum19/vcf_processing_analysis/tree/main



Serialize VCF file into
RDF (ttl)

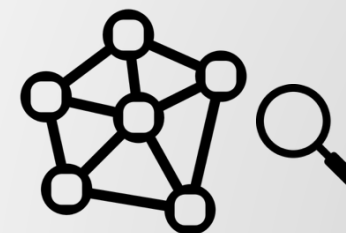
Query:

SELECT ?o

WHERE {

id:MutantID info:Variant ?o .

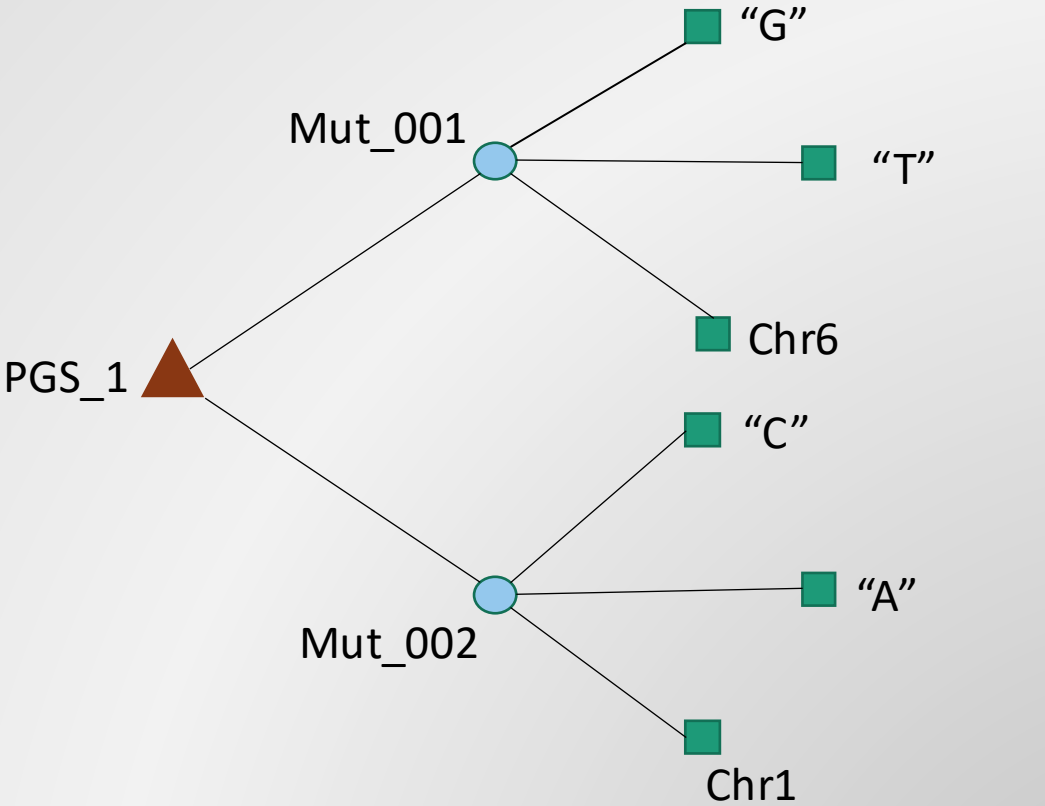
}



SPARQL query using
RDFLib

Potential Approach *a.*

* With permission



For whole VCF file

■ = literal

○ = internationalized resource identifier (IRI)

Mut = mutation

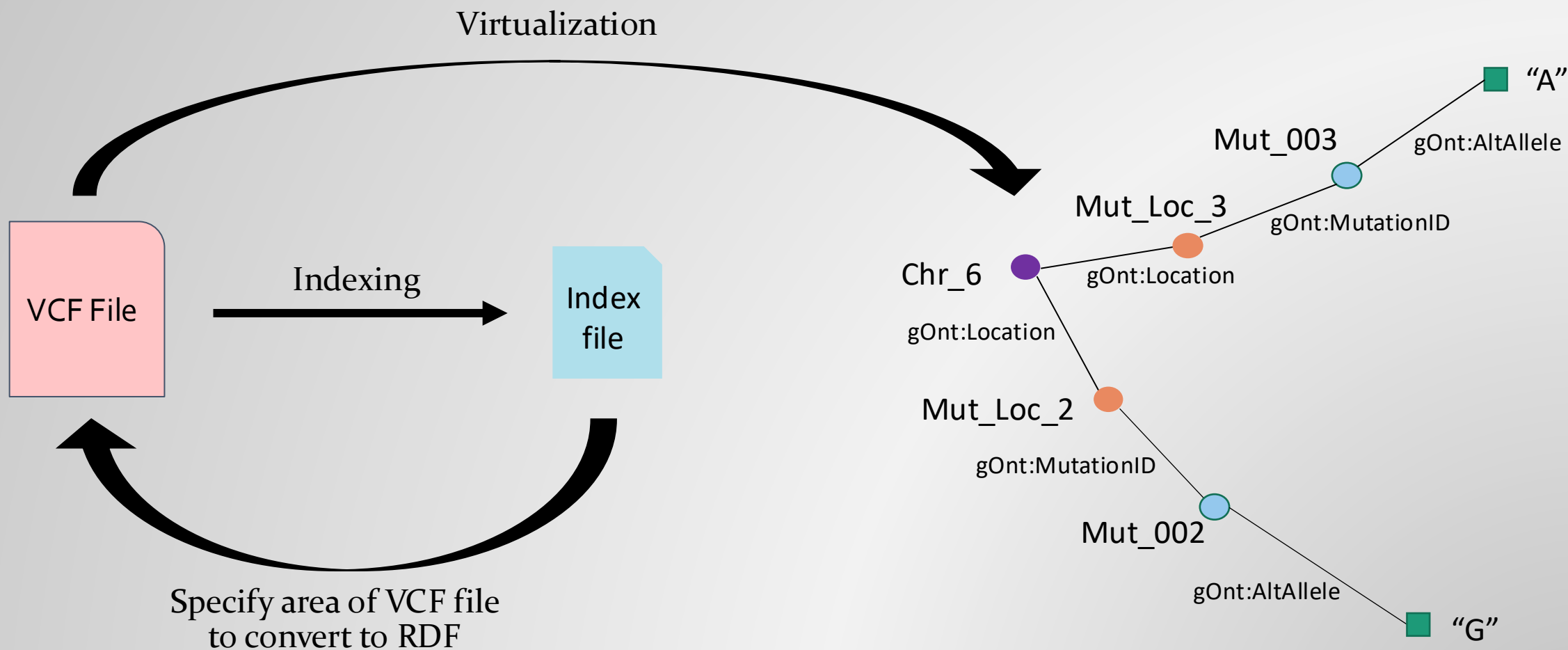
Potential Approach *b*.

■ = literal

● = internationalized resource identifier (IRI)

▲ = Graph name

* With permission





Questions?